EXPLORER

MACHINELEARNING

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exercise1.py u

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```python
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression

data = fetch_california_housing()
X = data.data
y = data.target
feature_names = data.feature_names

X10 = X[::10]
y10 = y[::10]

model = LinearRegression()
model.fit(X10, y10)

print("Features:", feature_names)
print("Coefficients:", model.coef_)
print("Intercept:", model.intercept_)

x_new = np.array([[8.3153, 41.0, 6.894423, 1.053714, 323.0, 2.533576, 37.88, -122.23]])
pred = model.predict(x_new)[0]
print("Prediction (Median House Value):", pred)
```

PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS

```
/usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_4/WilliamsBenjamin_Lab4/Exercise1/exercise1.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_4/WilliamsBenjamin_Lab4/Exercise1/exercise1.py
Features: ['MedInc', 'HouseAge', 'AveRooms', 'AveBedrms', 'Population', 'AveOccup', 'Latitude', 'Longitude']
Coefficients: [ 4.45792610e-01  1.01555473e-02 -1.32921383e-01  7.72819807e-01
  1.27153353e-05 -1.19226428e-01 -4.14933845e-01 -4.23452261e-01]
Intercept: -35.57248133900797
Prediction (Median House Value): 4.191623804793885
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$
```

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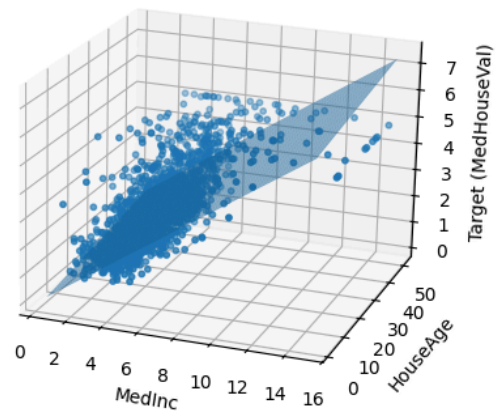
```python
import numpy as np
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing
from sklearn.linear_model import LinearRegression

data = fetch_california_housing()
X = data.data
y = data.target
feature_names = data.feature_names

X2 = X[::10, :2]
y2 = y[::10]

model = LinearRegression()
model.fit(X2, y2)

x0 = X2[:, 0]
x1 = X2[:, 1]

x0_grid, x1_grid = np.meshgrid(
    np.linspace(x0.min(), x0.max(), 40),
    np.linspace(x1.min(), x1.max(), 40)
)
grid = np.c_[x0_grid.ravel(), x1_grid.ravel()]
y_grid = model.predict(grid).reshape(x0_grid.shape)

fig = plt.figure()
ax = fig.add_subplot(111, projection="3d")

ax.plot_surface(x0_grid, x1_grid, y_grid, alpha=0.55)
ax.scatter(x0, x1, y2, s=10)

ax.set_xlabel(feature_names[0])
ax.set_ylabel(feature_names[1])
ax.set_zlabel("Target (MedHouseVal)")
ax.set_title("Multiple Linear Regression (2 features): Plane + Scatter")

plt.show()
```

Figure 1



x=3.4318, y pane=55.7188, z=5.2304

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```python
import numpy as np
from sklearn.datasets import fetch_california_housing
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LinearRegression

data = fetch_california_housing()
X = data.data
y = data.target
feature_names = np.array(data.feature_names)

scaler = StandardScaler()
Xz = scaler.fit_transform(X)

model = LinearRegression()
model.fit(Xz, y)

coefs = model.coef_
idx = np.argmax(np.abs(coefs))

print("All coefficients (standardized X):")
for n, c in zip(feature_names, coefs):
    print(n, c)

print("\nMost-weight coefficient (by absolute value):")
print(feature_names[idx], coefs[idx])
```

PROBLEMS | OUTPUT | DEBUG CONSOLE | TERMINAL | PORTS

```
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_4/WilliamsBenjamin_Lab4/Exercise3/exercise3.py
All coefficients (standardized X):
MedInc 0.82961930428045
HouseAge 0.11875165121214185
AveRooms -0.26552687950662
AveBedrms 0.30569622980430844
Population -0.0045030013126143055
AveOccup -0.039326266978148824
Latitude -0.8998856544145066
Longitude -0.8705410023357308

Most-weight coefficient (by absolute value):
Latitude -0.8998856544145066
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$
```

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CHAT

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Figure 1
MedInc
HouseAge
AveRooms
AveBedrms
Population
AveOccup
0
5
10
15
20
40
50
100
30
10000
20000
30000
500
1000

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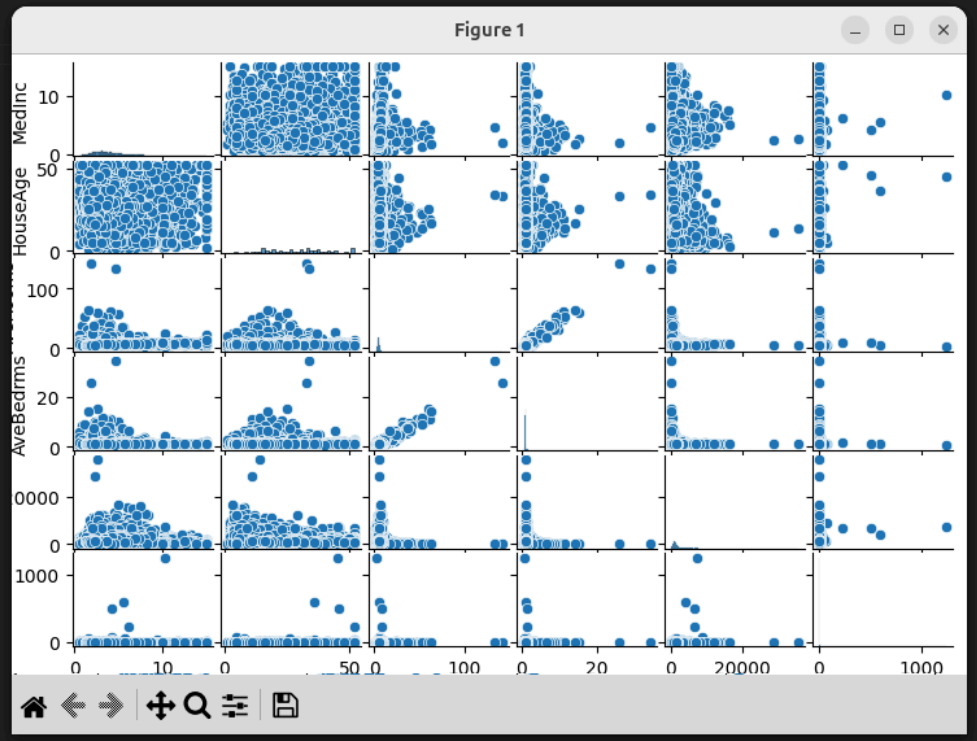
```python
import pandas as pd
import seaborn as sns
import matplotlib.pyplot as plt
from sklearn.datasets import fetch_california_housing

data = fetch_california_housing()
df = pd.DataFrame(data.data, columns=data.feature_names)

df2 = df.drop(columns=["Longitude", "Latitude"])

sns.pairplot(df2, diag_kind="hist")
plt.show()
```



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```
/usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_4/WilliamsBenjamin_Lab4/Exercise4/exercise4.py
ben@ben-System-Product-Name:~/Documents/Development/MachineLearning$ /usr/local/bin/python3.14 /home/ben/Documents/Development/MachineLearning/Lab_4/WilliamsBenjamin_Lab4/Exercise4/exercise4.py
```

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python3.14

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